From Raw Shots to Expectation Values: Shot Grouping Improves Time Series Prediction

Markus Baumann*, Maximilian Zorn*, Thomas Gabor†, Claudia Linnhoff-Popien*, and Jonas Stein*
*QAR-Lab, Department of Computer Science, LMU Munich, Munich, Germany
†Department of Computer Science, University of Exeter, Exeter, United Kingdom

***Abstract*—Between keeping every shot as a noisy training example and averaging all shots into one clean feature lies an entire spectrum of data organizations that quantum machine learning usually leaves implicit. We make that choice explicit. Shot grouping partitions a fixed measurement record into disjoint groups and averages within each group, tuning smoothly between the two conventions with a single validated parameter and no additional quantum executions. We evaluate the method on chaotic synthetic benchmarks and on real-world data under strict execution budgets. An intermediate grouping consistently outperforms the standard expectation-value pipeline. Matched controls show that the gain acts as correctly scaled regularization of the classical readout. Unmitigated runs on two superconducting processors confirm the trend on every task. Shot grouping therefore offers a practical way to improve finite-shot quantum learning at no added quantum cost.**

***Index Terms*—Quantum computing, Machine learning, Reservoir computing, Time series analysis**

I. Introduction

Every quantum learning experiment ends the same way: a circuit runs many times, and each run returns one measured bitstring. Before anything can be learned, this pile of shots must become training data. The step looks like bookkeeping, and it is usually treated that way. Almost every pipeline averages all shots of a time step into a single expectation-value estimate and moves on. This paper asks what that habit costs.

The habit has a respectable origin. Expectation values are the native currency of quantum mechanics: they are what theory predicts and what experiments are built to estimate. A learning pipeline, however, does not consume observables. It consumes training examples, and it cares about two things only: how many there are and how noisy each one is. The moment a measurement record becomes training data, a physics convention quietly turns into a statistical decision: spend all measurements on making one example clean, or make several examples that are merely clean enough. No single choice can suit every task, budget, and noise level.

Averaging everything is one end of a dial. It buys the cleanest possible feature vector but pays with scarcity. Thousands of circuit executions collapse into a handful of training examples, and too few examples relative to the number of features destabilizes regression [1]. The opposite end, keeping every shot as its own example, buys abundance but pays with noise, and noisy features systematically shrink fitted coefficients [2]. Between these extremes lies an entire range of aggregations.

Corresponding author: markus.baumann@campus.lmu.de.

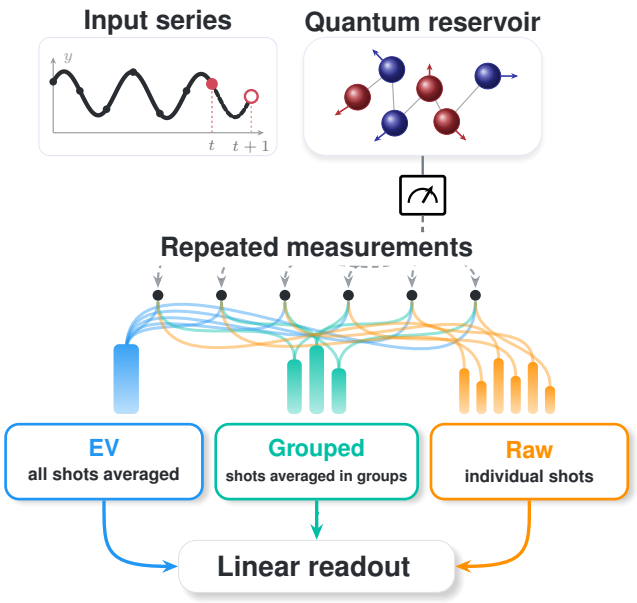


Fig. 1. The one idea of this paper, in a picture. A normalized input drives a fixed quantum reservoir that is measured many times at each step, which produces a finite record of shots (black dots). Turning that record into training data is the only step that changes. EV averages all shots of a step into one clean feature. Raw keeps every shot as its own noisy feature. Grouped sits between them and averages the shots in disjoint groups, giving a few partly cleaned features. All three feed the same linear readout, and the circuit, the shots, and the targets are identical across them, so any difference in accuracy comes from the aggregation alone.

Nothing in the record forces either convention, and the record does not care which one tradition prefers.

Quantum reservoir computing (QRC) is a clean place to study this dial [3]–[5]. A fixed quantum circuit transforms an input sequence into measured features, and only a classical readout is trained. The circuit, the targets, and the measurement record can therefore be held completely fixed while the aggregation alone changes. Whatever happens to the forecast is attributable to the aggregation because nothing else moved.

We compare three representations of one record (Fig. 1). The expectation-value representation (EV) averages all measurements at each time step. Raw uses each measurement separately. Grouped averages disjoint subsets, and its aggregation scale k, the number of shots per averaged training view, slides smoothly from $k = 1$ (Raw) to $k = N$ (EV).

The dial turns out to matter. Under one matched protocol and fixed execution budgets, an intermediate aggregation lowers test error relative to full averaging on every task we evaluate,

including two synthetic systems and two real-world datasets. The best aggregation scale depends on the record, so no single convention is appropriate. The gain requires nothing quantum. Matched controls trace it to correctly scaled regularization of the classical readout, supplied automatically by the record's own fluctuations. Because aggregation is classical post-processing, it costs no additional quantum executions and applies retroactively to any archived record.

Our contribution is *adaptive shot aggregation*. The number of shots per training view becomes a setting of the classical readout, selected by validation like any other hyperparameter, at no additional quantum cost. We map when the choice matters through three budget allocations of time steps and shots. We explain why it works through matched controls that isolate the regularization mechanism. We test where it transfers through unmitigated experiments with a separate hardware reservoir on two superconducting processors.

II. Related Work

Finite sampling can be addressed at three points in a quantum learning pipeline. It can be addressed before execution, by deciding how to spend a measurement budget. It can be addressed after averaging, by correcting the statistics of expectation values. Or it can be addressed at the record itself, by changing how repeated measurements become training data. Our method lives at this third point, which is the least studied of the three.

The idea of averaging subsets of measurements is not new, and we do not claim it. Cemin *et al.* average subsets of measurements to enlarge a training set for reduced quantum channel learning [6], and it is a classical result that training with noisy inputs behaves like an additional penalty on large fitted weights [7]. What has been missing is the treatment of the aggregation scale as a tunable quantity. This study evaluates the complete range from Raw to EV on one fixed QRC record, calibrates the scale jointly with the readout regularization, gives an exact account of the penalty this creates, and tests the result under three allocations of a fixed execution budget on four matched forecasting tasks. The contribution is the calibration and evaluation of the aggregation scale, not the original introduction of subset averaging.

The other two points of the pipeline are well developed and complementary to ours. Before execution, resource-aware optimization decides how many measurements each part of a computation deserves [8]. After averaging, Ahmed *et al.* derive noise-aware training rules for expectation values [9], and Kreplin and Roth reduce sampling noise through variance regularization [10]. Fundamental limits of learning from finitely sampled physical systems have also been mapped out [11]. None of these methods competes with ours. Allocation ends where our method begins, and the averaged quantities that noise-aware corrections operate on can themselves be produced by grouped aggregation. Our method needs nothing beyond a single archived measurement record.

Within QRC, a rich line of work studies how noise, dissipation, measurement, finite Hilbert space, and input transformations shape temporal features [12]–[23]. All of these studies analyze or modify the quantum representation. We do the opposite. The representation is held completely fixed, and only the organization of its repeated measurements changes. Because every reservoir in those studies still produces a measurement record, adaptive shot aggregation applies on top of any of them.

III. Method

A. Adaptive shot aggregation

Write $x_t^{(n)}$ for the features produced by shot n at time t. We split the N shots of each step into $G = N/k$ groups of equal size k and average within each group,

$$x_t^{(g)} = \frac{1}{k} \sum_{n \in g} x_t^{(n)}, \qquad g = 1, \ldots, G. \tag{1}$$

Every group is a partly averaged view of the same time step, and all views share the same target y_t. At test time we average the G predictions of a step, which is identical to predicting once from the averaged features. The two methods are therefore scored in exactly the same way, so any difference between EV and Grouped is created during training, not at test time.

The shot index carries no physical meaning, because shot n at one step and shot n at the next come from unrelated circuit runs. We therefore shuffle the shots freshly at every time step before forming groups, and we repeat the whole analysis over several random groupings.

B. Fixed execution budget and training

Evaluating T time steps with N shots per step costs TN circuit executions. We hold this budget fixed at ten thousand executions and study three ways of spending it, $(T, N) = (250, 40)$, $(500, 20)$, and $(1000, 10)$, ranging from fewer time steps with more shots to more time steps with fewer shots. For each setting we try every group size k that divides N, which includes Raw at $k = 1$ and EV at $k = N$.

Using three expanding splits, each candidate pair of group size and ridge strength (k, λ) is fitted on an earlier block and scored on the block that follows. The best pair is then frozen, the readout is refitted on all data before the test block, and the final fifth of the series is scored once. Normalization and feature scaling use training data only, so the test block never influences any choice.

C. Why grouping regularizes the readout

Full averaging hands the readout one clean picture of each time step. Individual shots hand it many pictures, each blurry. Intermediate groups hand it a few lightly cleaned pictures of the same target, and the differences between those pictures reveal which parts of the feature vector are stable and which are only measurement noise. Fitting the readout to all of the views at once discourages it from trusting the unstable parts.

For a linear readout with squared loss this becomes an exact identity. Let $\bar{x}_t$ be the average of the G grouped views, and let $C_{k,t} = \frac{1}{G} \sum_g (x_t^{(g)} - \bar{x}_t)(x_t^{(g)} - \bar{x}_t)^\top$ record how those views scatter around their average, that is, how the features wobble

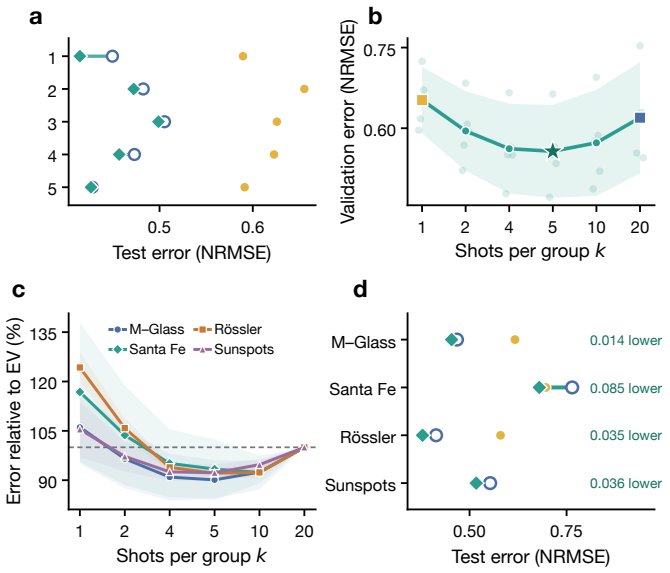


Fig. 2. How adaptive aggregation behaves under the shared protocol, read left to right. Lower NRMSE is always better. (a) One task, five runs. For each of five independent Mackey–Glass trajectories, Grouped (teal diamond) lands to the left of full averaging EV (open circle), and individual shots Raw (yellow) sit farthest right. Grouped is the most accurate on every run and Raw the least. (b) Turning the dial. At the balanced budget we sweep the aggregation scale k from Raw at $k = 1$ (yellow square) to EV at $k = 20$ (blue square). The error traces a U shape whose lowest point is an intermediate scale, which is where Grouped operates. The star marks the scale chosen automatically by validation. (c) The U shape is general. Plotting each task's error as a percentage of its own EV error, every curve dips below 100 at intermediate k, so an intermediate scale beats full averaging on all four tasks. (d) The bottom line. After k and the ridge strength are chosen on earlier data, Grouped (teal diamond) again beats EV (open circle) on every task by the amount shown at right, with Raw (yellow) trailing. Panels (b) and (c) use validation data only. Panels (a) and (d) use held-out test blocks. Bands and intervals are computed across independent trajectories or windows, and faint marks show repeated measurement trials within them.

together from group to group. For any intercept b and weights w,

$$\frac{1}{G}\sum_{g}\left(y_t - b - w^\top x_t^{(g)}\right)^2 = \left(y_t - b - w^\top \bar{x}_t\right)^2 + w^\top C_{k,t}\, w. \quad (2)$$

The first term on the right is exactly the EV loss. The second charges extra whenever the weights lean on feature combinations that vary strongly across the views. Grouped training is therefore EV training plus a penalty read directly off the observed shot fluctuations, and the identity assumes neither Gaussian noise nor independence over time.

The identity motivates a matched baseline: covariance-regularized expectation-value training (Cov-EV). Cov-EV keeps the fully averaged EV features and adds a penalty estimated from how the individual shots fluctuate together. Its strength and the ordinary ridge strength are chosen using the same earlier data as the other methods. Cov-EV does not reuse the aggregation scale chosen by Grouped; it tests whether the same regularization mechanism can be applied directly after full averaging. Similar performance from Grouped and Cov-EV would therefore support regularization as the source of the gain.

The equivalence is also verified computationally. When k is fixed, Grouped training and EV training with the penalty defined by $C_{k,t}$ give identical predictions. Because the leaky integrator and overlapping lags make neighboring feature vectors dependent, we report uncertainty across trajectories or measured windows rather than across the many correlated grouped views.

IV. Experimental Setup

We run two studies. The first is a controlled simulation on four forecasting tasks. The second repeats the core comparison on real quantum hardware. In both, every aggregation choice reads the very same measurement record, so any difference in accuracy comes only from how the shots are organized and not from running the circuit differently.

A. Simulation reservoir and setup

The simulation reservoir is a fixed six-qubit circuit with two shallow data reuploading layers [24]. Each layer applies input-dependent R_y and R_z rotations with fixed per-qubit offsets, neighboring R_{zz} couplings, and fixed non-Clifford mixing rotations. All angles are drawn once and then frozen; additional frozen circuits measure sensitivity to that draw. Each computational-basis measurement provides six local signals Z_i and five neighboring correlations Z_iZ_{i+1}. A leaky integrator with leak 0.2 and the ten most recent steps provide temporal memory, and ridge regression maps these features to the next value.

We forecast four time series one step ahead. Two are chaotic synthetic systems, Mackey–Glass [25] and Rössler [26]. Two are measured real-world series, the Santa Fe laser intensity [27] and the monthly sunspot number [28]. All four pass through this frozen circuit and the identical processing pipeline, with the same features, memory, scaling, model selection, and budgets. The only thing that changes from task to task is the input sequence. This isolates the effect of aggregation from any per-task tuning.

Every method reads the same raw record. Next to EV, Grouped, and Raw we include three references that separate the aggregation effect from the quantum device. Cov-EV adds a shot-covariance penalty to plain averaging, an ideal noise-free version of the reservoir removes sampling noise entirely, and the classical input processed in the same way removes the reservoir. To keep the statistics honest, each synthetic task is run on five independently generated trajectories and each real series on five separate time windows with disjoint test blocks. Within each of these units we repeat the experiment over three circuit settings, three measurement draws, and three random groupings, average those repeats, and only then form paired confidence intervals across the five independent units. Repeated simulations of one sequence are therefore never counted as independent forecasting problems.

Model selection never touches the test data. After a short washout, the final fifth of each series is set aside and left untouched while settings are chosen. The earlier part is used in three expanding rounds that always fit on the past and check on the block that follows, and only after the group size and ridge strength are fixed is the reserved fifth scored once. Grouped predictions are averaged back to one forecast per step, so every

method is judged on the same forecasts. The main budget places 20 shots at each of 500 time steps. Two further ways of spending the same ten thousand executions, more shots with fewer steps and fewer shots with more steps, serve as secondary analyses. All reservoir parameters, data generators, window origins, and selection rules are fixed and stored before any test block is seen.

B. Hardware setup

The hardware study deliberately uses a different reservoir. It is a fixed ten-qubit circuit with one input qubit, nine memory qubits, a single ring of entangling gates, and ten local outputs, mapped to selected qubits on the *ibm_fez* and *ibm_marrakesh* processors and held fixed across all four tasks. Its purpose is to check whether the same aggregation procedure still helps on a second circuit and a different set of measured features. It is not a hardware implementation of the six-qubit simulator and not a comparison of qubit counts. As in simulation, every method receives the same circuit outputs and only the classical grouping of shots changes. For each task and processor we take one acquisition of 400 shots at each of 500 time steps and split it into twenty ordered, disjoint records of 20 shots. The resulting intervals describe how much the result varies among the measurement blocks of that single acquisition. They do not stand in for independent devices or separate calibration periods.

V. Evaluation

We report accuracy as test NRMSE, a normalized error for which lower is better and where one means predicting the mean. Three ways of turning the shot record into features are compared throughout. EV averages all shots of a step into one clean feature vector. Raw keeps every shot as its own noisy example. Grouped is our method and sits between them. A fourth entry, Cov-EV, is a control that adds an independently tuned covariance penalty to plain averaging. We summarize the benefit of Grouped as the paired difference EV minus Grouped, so a positive value means Grouped is more accurate. Each difference comes with a 95% confidence interval. An interval above zero provides evidence of a positive mean improvement; an interval containing zero is inconclusive. The balanced setting spends the fixed budget on 500 time steps with 20 shots each.

A. Balanced setting

At the balanced budget, Grouped lowers mean NRMSE relative to EV on all four tasks (Table I, Fig. 2). The improvement is 0.0138 on Mackey–Glass (95% CI $[-0.0023, 0.0300]$), 0.0848 on Santa Fe laser ($[0.0174, 0.1522]$), 0.0349 on Rössler ($[0.0139, 0.0559]$), and 0.0360 on sunspots ($[0.0103, 0.0616]$). The interval stays above zero on the two real-world datasets, Santa Fe laser and sunspots, and on Rössler. Only Mackey–Glass is inconclusive.

Santa Fe laser shows the largest effect. Grouped reduces the error from 0.764 to 0.680, and all five disjoint test windows favor Grouped. When k and the ridge strength are tuned together, the search lands on $k = 10$ most often and on an intermediate scale, neither Raw nor EV, in 94% of repeated trials. Cov-EV reaches 0.677, almost the same as Grouped, which already suggests that the gain is a regularization effect rather than new physical information. The gain also depends on how the budget is spent. On Santa Fe it shrinks from 0.143 with 40 shots per step to 0.010 with 10 shots per step because grouping has more shots to reorganize when shots are plentiful.

TABLE I
Mean test NRMSE at the balanced setting. Repeated measurement trials are averaged within each of five separate trajectories or windows. Lower values are better.

Task	Class.	Ideal	EV	Grouped	Raw	Cov-EV
Mackey–Glass	0.100	0.178	0.468	0.454	0.617	**0.447**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

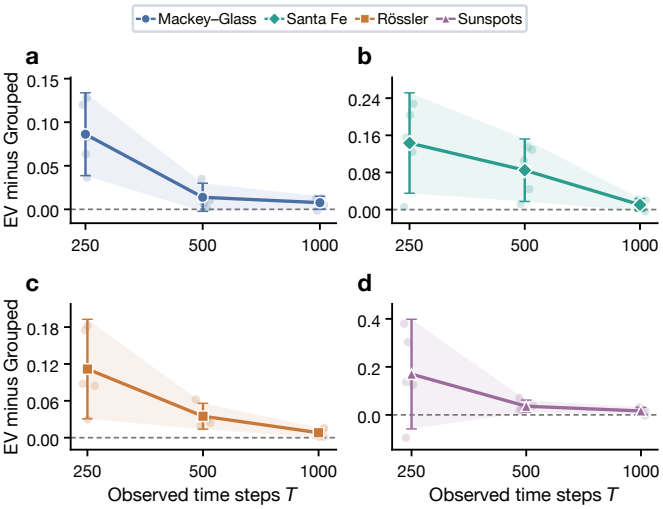


Fig. 3. When aggregation helps most, at a fixed budget of ten thousand circuit executions. The vertical axis shows how much Grouped improves on full averaging, measured as the EV error minus the Grouped error, so a value above the dashed zero line means Grouped wins and a value on the line means a tie. Moving right spends the same budget on more time steps and therefore fewer shots at each step, from 40 shots at 250 steps to 10 shots at 1000 steps. In every task the improvement is largest on the left and shrinks toward zero on the right. The message is simple. Grouping helps most when each step is measured many times and the series is short, because that is exactly when full averaging wastes the most training examples. Points are trajectory or window units, and markers with bars are means with 95% intervals.

We also compare an ideal noise-free reservoir with a classical baseline that uses identical preprocessing. The quantum reservoir is worse on Mackey–Glass and sunspots, better on Santa Fe, and inconclusive on Rössler. Our claims are therefore about organizing a chosen reservoir record, not quantum advantage. The preferred aggregation scale varies across tasks, trajectories, circuit settings, measurement draws, and grouping assignments, and EV is occasionally best. Equation (2) explains this variation: changing k changes which feature combinations the readout is discouraged from using, not merely the strength of that regularization. Raw is less accurate than Grouped on all four task means.

B. Trading time steps against shots

Figure 3 compares three ways of spending the fixed budget. With 40 shots at each of 250 time steps, Grouped improves

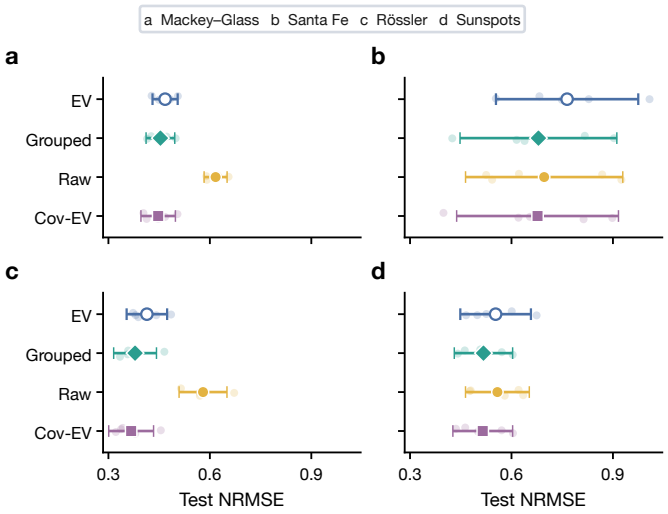


Fig. 4. Why the gain is regularization, shown task by task. Each panel places the four methods on the same error axis, where lower is better. Grouped (teal diamond) and Cov-EV (purple square) sit lowest and almost on top of each other on every task, EV (open circle) is behind them, and Raw (yellow, individual shots) is worst. The near-match between Grouped and Cov-EV is the key point. Cov-EV adds Grouped's fluctuation penalty directly to full averaging, so when it reaches the same error the benefit of grouping is regularization of the readout rather than extra information from the device. Faint points are the five trajectory or window means after repeated measurement trials are averaged within each unit. Markers and bars show the means and their 95% intervals. The order from top to bottom is EV, Grouped, Raw, and Cov-EV.

every task by 0.086 to 0.170. The interval stays above zero for Mackey–Glass, Santa Fe laser, and Rössler, while the sunspot result varies across windows. When the same budget is spread over 1000 time steps with only 10 shots each, the gains shrink to between 0.008 and 0.016, and the interval stays above zero only for Mackey–Glass and sunspots.

C. Why the gain is regularization

Cov-EV keeps EV features and adds a penalty based on how the shots fluctuate together. At the balanced setting its mean error is 0.0021 to 0.0116 NRMSE below Grouped, and only the Rössler interval excludes zero. This near-match supports the regularization mechanism in Eq. (2) rather than extra device information (Fig. 4). Raw is less accurate than EV on Mackey–Glass and Rössler and inconclusive on the two real-world datasets. At fixed k, the exact two-term loss reproduces the Grouped ridge fit to numerical precision; Cov-EV independently tunes the strength of the full shot-covariance penalty.

Grouping is also robust to how the groups are drawn. Across twenty repeated settings the average improvements are -0.0064 on Mackey–Glass, 0.1103 on Santa Fe laser, 0.0177 on Rössler, and 0.0493 on sunspots. Only the sunspot interval stays above zero, reflecting the wide variation across settings. Fixing one grouping across time instead of reshuffling at every step changes the error by at most 0.0021 NRMSE, with every interval containing zero. The grouping convention therefore matters far less than ordinary run-to-run variation. Figure 4 collects these controls.

D. Transfer to hardware

Figure 5 reports the unmitigated hardware results. Grouped improves every task, with mean gains from 0.038 to 0.153 NRMSE on *ibm_fez* and from 0.031 to 0.103 on *ibm_marrakesh*. On Santa Fe laser, the error falls from 0.750 to 0.711 on *ibm_fez* and from 0.744 to 0.713 on *ibm_marrakesh*. Grouped is more accurate in 19 of 20 measurement blocks on each processor.

VI. Discussion

A measurement record contains training structure that full expectation-value averaging discards. Choosing the aggregation scale on earlier time blocks recovers part of it without further quantum executions. Adaptive shot aggregation changes only classical processing: k moves from Raw at $k = 1$ to EV at $k = N$. The scale and ridge strength are fixed before the held-out test; the circuit and record remain unchanged.

The method helps most in a clear regime. When a record holds many shots at each of relatively few time steps, full averaging turns a large measurement effort into only a handful of training examples. Partial averaging spends those same shots differently. It produces several lightly denoised views of each target, and the small differences among the views act as a penalty that discourages the readout from trusting feature directions dominated by measurement noise. Equation (2) makes this exact. Grouped training is ordinary averaging plus a penalty read from the observed shot fluctuations.

Our control confirms this reading. Cov-EV applies the same type of covariance penalty after full averaging and reaches

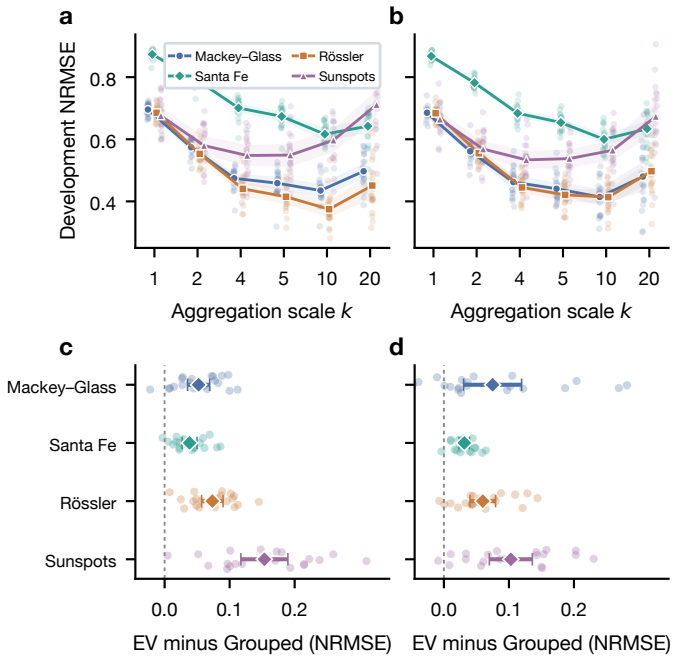


Fig. 5. The same effect on real hardware, on two processors. A separate ten-qubit reservoir was run without error mitigation on *ibm_fez* and *ibm_marrakesh*. For each task and processor, one acquisition of 400 shots at each of $T = 500$ time steps was split into twenty disjoint records with $N = 20$. Panels (a,c) show *ibm_fez*; panels (b,d) show *ibm_marrakesh*. (a,b) The aggregation spectrum. On both devices the development error traces the same U shape over the aggregation scale k, so an intermediate scale again beats both endpoints on every task. (c,d) The improvement per task, measured as the EV error minus the Grouped error, where the dashed line is a tie and points to its right favor Grouped. Grouped is more accurate on all four tasks on both machines. Faint points are the individual measurement blocks, and each colored marker with a bar is the mean with its 95% interval. These intervals describe variation among the blocks of a single acquisition, not across independent hardware campaigns.

slightly lower mean error, yet stays close to Grouped on every task. Their similar performance supports covariance regularization as a shared mechanism. The small gap shows that grouping is one convenient way to obtain this regularization, not the only one, and that the benefit arises in the classical fit rather than from additional physical information.

The effect also transfers. It shows up in the six-qubit simulation reservoir and again in a separate ten-qubit circuit run on two processors.

A practical consequence follows. A different group size can only be built from the individual shot outcomes, so trying other scales later is possible only if those outcomes were kept. An experiment that stores just the averaged expectation value has already thrown away the option to regroup, while an experiment that keeps its full shot record leaves every group size open for free.

The hardware results have a deliberately narrow scope. Their confidence intervals are computed from the measurement blocks of one acquisition on each task and processor. They therefore show that the effect is stable within a single run, not that it repeats across separate runs on different days or calibration settings. Confirming that reproducibility is the main open question, and the natural next tests are independent hardware campaigns, other reservoir families, and richer readouts.

VII. Conclusion

Turning a shot record into training data has always involved a choice, and the field has answered it by reflex, averaging everything. We show the reflex is beatable. Choosing the aggregation scale by validation lowers mean test error on chaotic and real-world forecasting at the balanced budget, holds on two superconducting processors, and requires no additional quantum cost. Because every measurement-based reservoir produces such a record, the choice we make explicit here is available to all of them. And the underlying question, how to turn finitely sampled noisy measurements into training data, reaches well beyond reservoirs to finite-shot learning as a whole.

References

[1] D. Hsu, S. M. Kakade, and T. Zhang, "Random Design Analysis of Ridge Regression," in *Proceedings of the 25th Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 23, 2012, pp. 9.1–9.24.

[2] W. A. Fuller, *Measurement Error Models*. New York: Wiley, 1987.

[3] K. Fujii and K. Nakajima, "Harnessing Disordered-Ensemble Quantum Dynamics for Machine Learning," *Physical Review Applied*, vol. 8, no. 2, p. 024030, 2017.

[4] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, "Boosting Computational Power through Spatial Multiplexing in Quantum Reservoir Computing," *Physical Review Applied*, vol. 11, no. 3, p. 034021, 2019.

[5] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi *et al.*, "Opportunities in Quantum Reservoir Computing and Extreme Learning Machines," *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.

[6] G. Cemin, M. Cech, E. Weiss, S. Soltan, D. Braun *et al.*, "Machine Learning of Reduced Quantum Channels on Noisy Intermediate-Scale Quantum Devices," *Physical Review A*, vol. 110, no. 5, p. 052418, 2024.

[7] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[8] C. Moussa, M. H. Gordon, M. Baczyk, M. Cerezo, L. Cincio, and P. J. Coles, "Resource Frugal Optimizer for Quantum Machine Learning," *Quantum Science and Technology*, vol. 8, no. 4, p. 045019, 2023.

[9] O. Ahmed, F. Tennie, and L. Magri, "Optimal Training of Finitely Sampled Quantum Reservoir Computers for Forecasting of Chaotic Dynamics," *Quantum Machine Intelligence*, vol. 7, p. 31, 2025.

[10] D. A. Kreplin and M. Roth, "Reduction of Finite Sampling Noise in Quantum Neural Networks," *Quantum*, vol. 8, p. 1385, 2024.

[11] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci *et al.*, "Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks," *Physical Review X*, vol. 13, no. 4, p. 041020, 2023.

[12] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Quantum Noise-Induced Reservoir Computing," 2022.

[13] A. Palacios de Luis, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Role of Coherence in Many-Body Quantum Reservoir Computing," *Communications Physics*, vol. 7, no. 1, p. 369, 2024.

[14] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Temporal Information Processing Induced by Quantum Noise," *Physical Review Research*, vol. 5, no. 2, p. 023057, 2023.

[15] R. Martínez-Peña and J.-P. Ortega, "Quantum Reservoir Computing in Finite Dimensions," *Physical Review E*, vol. 107, no. 3, p. 035306, 2023.

[16] P. Mujal, R. Martínez-Peña, G. L. Giorgi, M. C. Soriano, and R. Zambrini, "Time-Series Quantum Reservoir Computing with Weak and Projective Measurements," *npj Quantum Information*, vol. 9, p. 16, 2023.

[17] Y. Suzuki, Q. Gao, K. C. Pradel, K. Yasuoka, and N. Yamamoto, "Natural Quantum Reservoir Computing for Temporal Information Processing," *Scientific Reports*, vol. 12, p. 1353, 2022.

[18] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Dissipation as a Resource for Quantum Reservoir Computing," *Quantum*, vol. 8, p. 1291, 2024.

[19] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, and T. A. Ohki, "Nonlinear Input Transformations Are Ubiquitous in Quantum Reservoir Computing," *Neuromorphic Computing and Engineering*, vol. 2, no. 1, p. 014008, 2022.

[20] R. Martínez-Peña, J. Nokkala, G. L. Giorgi, R. Zambrini, and M. C. Soriano, "Information Processing Capacity of Spin-Based Quantum Reservoir Computing Systems," *Cognitive Computation*, vol. 15, no. 5, pp. 1440–1451, 2023.

[21] W. D. Kalfus, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, T. A. Ohki, and L. C. G. Govia, "Hilbert Space as a Computational Resource in Reservoir Computing," *Physical Review Research*, vol. 4, no. 3, p. 033007, 2022.

[22] D. Fry, A. Deshmukh, S. Y.-C. Chen, V. Rastunkov, and V. Markov, "Optimizing Quantum Noise-Induced Reservoir Computing for Nonlinear and Chaotic Time Series Prediction," *Scientific Reports*, vol. 13, p. 19326, 2023.

[23] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, and T. A. Ohki, "Quantum Reservoir Computing with a Single Nonlinear Oscillator," *Physical Review Research*, vol. 3, no. 1, p. 013077, 2021.

[24] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.

[25] M. C. Mackey and L. Glass, "Oscillation and Chaos in Physiological Control Systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977.

[26] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, 1976.

[27] N. A. Gershenfeld and A. S. Weigend, "The Future of Time Series," Santa Fe Institute, Tech. Rep. 93-08-053, 1993.

[28] SILSO World Data Center, "Monthly mean total sunspot number V2.0," Royal Observatory of Belgium, Brussels, 2026, dataset SN_m_tot_V2.0, series 1749–2026, accessed July 2026, CC BY-NC 4.0.